Factors Influencing the Adoption and Effectiveness of Drone-Based Medical Transport: A Systematic Review

By

Charitha Balaka Venkatareddy

Student Id number: 201862489

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I certify that this dissertation/project is entirely my own work

Supervisor: Dr Martyn Bradley

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List of Abbreviations

Medical Abbreviations

- **AED** = Automated External Defibrillator
- **EMS** = Emergency Medical Services
- **OHCA** = Out-of-Hospital Cardiac Arrest
- **RBC** = Red Blood Cell
- **WHO** = World Health Organization

Aviation and Operational Abbreviations

- **BVLOS =** Beyond Visual Line of Sight
- **CAA =** Civil Aviation Authority
- **FAA =** Federal Aviation Administration
- **GNSS =** Global Navigation Satellite System
- **GPS =** Global Positioning System
- **ICAO =** International Civil Aviation Organisation
- **UAS =** Unmanned Aircraft System
- **UAV =** Unmanned Aerial Vehicle

Methodological Abbreviations

- **BSC** = Business Source Complete

- **DEF** = Data Extraction Form
- **MMAT** = Mixed Methods Appraisal Tool
- **PRISMA** = Preferred Reporting Items for Systematic Reviews and Meta-Analyses
- **SLR** = Systematic Literature Review
- **TAM** = Technology Acceptance Model
- **UTAUT** = Unified Theory of Acceptance and Use of Technology

Introduction

The technological dimension of modern health care logistics has been undergoing a paradigm shift due to the rapid evolution of Unmanned Aerial Vehicles (UAVs), typically known as drones. Historically, drones were only utilised for military surveillance but now emerged as a disruptive innovation in medical logistics, offering new approaches for the transportation of sensitive materials and improving the 'last mile'[1] of supply chains (Hiebert et al., 2020; Daud et al., 2022). By employing this approach, the theoretical rationale lies in its "distinct operating capacity" such as self-navigational abilities and rapid transit speeds to overcome terrestrial obstacles such as traffic congestion, degraded road infrastructure and territorial inaccessibility (Hiebert et al., 2020; Daud et al., 2022). Consequently, the use of drones in health care logistics can be viewed not only as a modern innovation but also a strategic asset with the capacity to transport blood products, diagnostic samples, vaccines, automated external defibrillators (AEDs) and recently, viable human organs (Claesson et al., 2017; Scalea et al., 2021).

The idea was driven by the imperative to reduce preventable mortality and improve health equity, prompting governments, healthcare organizations and global health entities to explore the integration of UAVs into critical care paths. Large-scale deployments in resource-constrained environments, particularly in Rwanda and Ghana have demonstrated that aerial logistics can improve accessibility and reliability by surpassing the limitations of conventional ground infrastructure (Dutta and Samantarai, 2024). However, such successful implementations have inspired the concept of "technological belief" and that drones are a universal solution to the global logistics challenges. This optimism often creates a confirmation bias where successful "demonstration flights" are combined with systemic readiness, concealing the complex logistical friction that emerges when operations must scale from single

[1] **Last mile** - A final step in supply-chain which transports goods from distribution centre to end customer.

flights to continuous, 24-hour service. Despite this optimism, the existing evidence of the integration of UAVs into healthcare practice remains varied, experimental and fragmented. Clinical studies validate the stability of robust payloads. However, existing knowledge regarding organ transport and operational reliability in challenging weather conditions often relies on simulations or controlled environments instead of routine practice.

This disconnection highlights a major gap in the literature. Although the volume of research is increasing, current reviews predominantly remain descriptive, focusing on specific use cases such as disaster response, without examining the structural barriers to routine adoption. Furthermore, theoretical frameworks such as Technology Acceptance Model (TAM) (Davis, 1989) propose that despite well-reflected benefits in terms of performance, health administrators intuitively label drones as disruptive or resource-intensive, thus creating a psychological barrier and inhibiting their adoption in health facilities. The hesitation is compounded by the contextual divergences, drivers for adoption in low-resource settings where drones are considered as an alternative for degraded infrastructure, differ fundamentally from high-resource settings focused on marginal efficiency gains. Consequently, effectiveness is seen as a context-dependent variable rather than a universal metric.

This systematic literature review addresses this gap by examining the question: ***What managerial, clinical, and operational factors influence the adoption and effectiveness of drone-based medical transport for time-critical healthcare items?*** Adopting a rigorous methodology, this review synthesises 37 core empirical sources and nine foundational theoretical works. The objective is to analyse the robustness and consistency of the evidence, identifying the various barriers that need to be addressed to promote safe, reliable, and scalable adoption.

Context

Drone delivery logistics is identified as a disruption in transport infrastructure driven by the advances in battery power and sensor technologies. However, the potential of drones in delivering health care by bypassing the terrestrial infrastructure is well-established in theory, their practical integration of drone delivery into healthcare logistics remains largely experimental and isolated. Most current deployments operate as pilot studies or feasibility assessments rather than sustained, budget-line operations. The transition from proof-of-concept to operational reality is uniquely restricted by the intersection of aviation law, medical governance, and digital infrastructure. This complexity necessitates a focused analysis of three distinct but interrelated domains: managerial, clinical, and operational factors.

The *Managerial* domain is crucial because healthcare adoption decisions are hardly informed and driven by technical capability alone. Healthcare decision makers usually face the challenge of navigating through a complex network of institutional readiness, cost structures, liability considerations, human capacity, and integration within current supply chains. Early literature suggests that, despite the strategic benefits, organisations are faced with issues regarding regulatory clarity and the "absorptive capacity" of their workforce (Stierlin et al., 2024; De Silvestri et al., 2023, World Health Organisation, 2022). This information on the management issues is essential in formulating adoption strategies that extend beyond short-term pilots funded by innovation grants.

The *Clinical* domain addresses the fundamental safety of transporting sensitive biological materials. Considering that biomedical payloads need to be transported in conditions that are highly stable, sterile, and at finely controlled temperatures, it is important to assess if the UAV transport preserves clinical quality. The high-frequency vibrations can cause haemolysis or shear stress, resulting in the potential degradation of the clinical effectiveness of

delivered medications. Although current literature suggests favourable results for robust payloads such as whole blood (Amukele et al., 2017), evidence remains insufficient regarding multi-organ transport, fragile biologics, and operations under challenging conditions. An analysis of clinical factors ensures that the review extends beyond technological optimism to address critical questions related to patient safety and clinical governance.

The *Operational* domain mainly focuses on the reliability, performance, and adaptability of drone-based systems. The most important factors to consider are flight stability, environmental constraints, battery capacity, airspace regulations, and technical failures. Operational reliability directly impacts the clinical outcomes like sample integrity, and managerial confidence. Accordingly, evaluating this aspect is essential for determining the ability of drones to consistently fulfil real-world healthcare demands (Hiebert et al., 2020; Quintanilla García et al., 2021).

Crucially, these three domains do not operate in a vacuum, they function as a mutually constraining ecosystem where failure in one area destabilises the entire logistics chain. For example, a drone flight that achieves operational success remains clinically ineffective if the vibration profile during transport compromises the integrity of a sensitive blood sample. A clinically certified transport protocol is practically ineffective if it cannot navigate the liability frameworks required for hospital insurance. Current literature often considers these dimensions as separate challenges, failing to address the interconnected effects that occur when operational volatility meets clinical rigidity. This gap between technical feasibility (*can it fly?*) and organisational readiness (*can it be integrated?*) represents an important limitation in current studies.

By incorporating the three domains, the research question offers an integrated perspective that explains the actual readiness for drone transport services. The justification for

this focus is strengthened by significant gaps in the existing literature. Many studies concentrate on distinct elements but lack an integrated framework to combine these fragmented insights. Very few reviews focus on methodological weaknesses or the interplay between these three factors. Even fewer provide critical narrative on the severe contextual differences between rural versus urban settings or low-resource versus high-resource healthcare systems. Therefore, this systematic review aims to synthesise and critically evaluate the fragmented evidence, offering a more comprehensive understanding of what enables or constrains the successful adoption of drone-based medical transport.

Methodology

Methodological Approach and Critical Reflection

This research adopts a Systematic Literature Review (SLR) methodology to examine the factors affecting drone-based medical transport. The selection of SLR instead of a traditional narrative review was a strategic decision based on the safety-critical aspects of healthcare logistics. Despite their flexibility, narrative reviews often show selection bias due to the lack of clear inclusion protocols required for validating evidence in crucial medical scenarios. The primary strength of the SLR methodology is its transparency and replicability, which are crucial for evidence-based management. This review follows a pre-defined protocol to reduce researcher bias, ensuring that the synthesis of findings on organ, blood, and diagnostic sample transport is based on credible empirical data rather than personal interpretation.

However, it is important to recognise the fundamental limitations of this methodological choice. The rigidity of the SLR process is defined by strict inclusion criteria and peer-reviewed limitations which require the exclusion of "grey literature," such as government white papers, industry reports, or preliminary technological assessments. In the rapidly evolving field of unmanned aerial vehicles (UAVs), industry reports often contain the most current operational data regarding flight hours and failure rates. The exclusion of non-peer-reviewed sources could omit important but less rigorous scientific findings on logistical challenges in real time. Although the SLR protocol reduced selection bias, restricting the search to English-language papers may exclude relevant drone pilot data from regions like East Asia and Latin America. Moreover, the methodology is resource-intensive, requiring rigorous cross-checking to guarantee validity. Despite these constraints, the choice remains justified, the medical transport of bio-sensitive materials requires a level of evidential certainty that can only be provided through a thorough and objective evaluation of empirical literature.

Data Sources and Search Syntax

A systematic search across five high-impact databases identified 186 references, with syntax tailored to each database to maximise sensitivity.

- **Business Source Complete (BSC)** contributed 54 papers using the field code TX, filtered for peer-reviewed journals (2015–2025).
- **Scopus** provided 53 references using TITLE-ABS-KEY syntax.
- **Emerald Insight** yielded 36 papers, applying Boolean operators to filter for operations and logistics journals.
- **Web of Science** resulted in 24 references using the TS= (Topic) operator to capture interdisciplinary studies.
- **PubMed** provided 19 records. The syntax explicitly used the [Title/Abstract] tag for every keyword (e.g., "medical logistics" [Title/Abstract]), filtering strictly for journals to avoid indexing non-logistics biomedical papers.

Table 1. Search strategy and initial results by database

Database	**Final Search String**	**Results**
Business Source Complete (EBSCO)	TX ("drone" OR "UAV" OR "unmanned aerial vehicle*" OR "medical drone*") AND TX ("healthcare" OR "hospital*" OR "medical logistics" OR "medical delivery" OR "blood transport") AND TX ("adoption" OR "implementation" OR "benefit*" OR "challenge*" OR "management" OR "organi*ational")	54

Emerald Insight	("drone" OR "UAV" OR "unmanned aerial vehicle*" OR "medical drone*") AND ("healthcare" OR "hospital*" OR "medical logistics" OR "medical delivery" OR "blood transport") AND ("adoption" OR "implementation" OR "benefit*" OR "challenge*" OR "management" OR "organi*ational")	36
PubMed	(("drone" [Title/Abstract] OR "UAV" [Title/Abstract] OR "unmanned aerial vehicle" [Title/Abstract] OR "medical drone" [Title/Abstract]) AND ("healthcare" [Title/Abstract] OR "hospital" [Title/Abstract] OR "medical logistics" [Title/Abstract] OR "medical delivery" [Title/Abstract] OR "blood transport" [Title/Abstract]) AND ("adoption" [Title/Abstract] OR "implementation" [Title/Abstract] OR "benefit" [Title/Abstract] OR "challenge" [Title/Abstract] OR "management" [Title/Abstract] OR "organi*ational" [Title/Abstract]))	19
Scopus	TITLE-ABS-KEY (("drone" OR "UAV" OR "unmanned aerial vehicle*" OR "medical drone*") AND ("healthcare" OR "hospital*" OR "medical logistics" OR "medical delivery" OR "blood transport") AND ("adoption" OR "implementation" OR "benefit*" OR "challenge*" OR "management" OR "organi*ational"))	53
Web of Science	TS= (("drone" OR "UAV" OR "unmanned aerial vehicle*" OR "medical drone*") AND ("healthcare" OR "hospital*" OR "medical logistics" OR "medical delivery" OR "blood transport") AND ("adoption" OR "implementation" OR "benefit*" OR "challenge*" OR "management" OR "organi*ational"))	24

Total		186

Study Selection: PRISMA

The selection method followed PRISMA 2020 guidelines to ensure replication of the process. Initially, 186 records were identified, from which duplicates were removed using EndNote and manual checking, which resulted a unique dataset for screening. Titles and abstracts were screened for relevance, and studies irrelevant to time-critical medical transport (e.g., drones for agricultural spraying or military surveillance) were excluded. A thorough review of the remaining publications was conducted according to strict eligibility criteria: English language, publication between 2015 and 2025, and specific focus on managerial, clinical, or operational factors of medical transport. The intensive screening process resulted in a total of 37 core empirical papers. Figure 1 illustrates the PRISMA flowchart, explaining the study selection process for this review analysis.

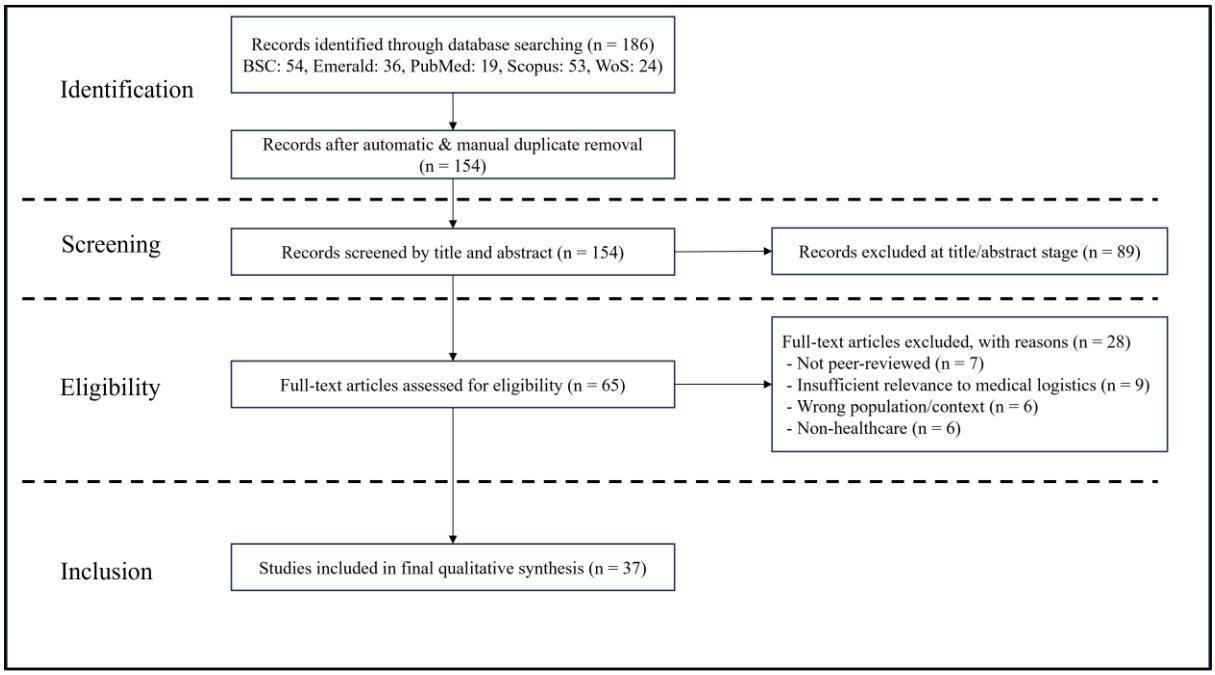


Figure 1. PRISMA flowchart for the relevant study selection

Quality Assessment (QA)

The Mixed Methods Appraisal Tool 2018 (MMAT 2018) has been utilised in this study for assessing the internal validity of the 37 core papers. The reasons why the MMAT 2018 was selected include the variability in the dataset studied in this research that ranges from quantitative simulation studies (Rees et al., 2021), perception studies in the case of qualitative research (Truog et al., 2020), and realistic case studies (Scalea et al., 2021).

Each paper was evaluated against five methodology-related criteria particularly to each study type (Appendix A). For simulation research papers with quantitative methods, the criteria involved the relevance of the simulation models and the clarity of variables. For the other type of papers with qualitative methods, the criteria involved credibility and confirmability.

Based on this analysis, each paper received a grade reflecting the weight of evidence in the synthesis (Appendix B):

- **High Quality (5/5 Criteria):** Studies that are more rigorously designed and documented (such as Amukele et al., 2017). This category of publications has been given maximum weightage in the clinical analysis.

- **Medium Quality (3-4/5 Criteria):** This category comprises studies that have minor flaws in methodology but significant information.

- **Low Quality (<3/5 Criteria):** Studies failing to meet the level of minimally required quality were excluded at the stage of full-text review to ensure that only sound methodological evidence was synthesized in the final synthesis to differentiate between those studies with a high level of internal validity, such as laboratory blood tests, and those with little or any ecological validity, such as theoretical routing models, a distinction that strengthens the critical synthesis in the Findings chapter.

The explicit grading ensures that the synthesis only took place through valid methodological evidence.

Selection of Theoretical Frameworks

This systematic review prioritises empirical evidence (2015-2025) to capture the rapid technological advancements of UAVs, while also incorporating additional foundational frameworks that predate this timeframe. The incorporation of these foundational works such as Davis (1989), Rogers (2003), and Venkatesh et al. (2003) is theoretically compelled by the descriptive nature of the 37 principal empirical studies. Although the core dataset provides robust data on ***what*** operational and clinical barriers exist, it largely lacks the theoretical mechanisms to explain ***why*** organisational adoption remains fragmented despite successful pilots. These older frameworks provide the necessary socio-technical foundation to interpret modern findings. For instance, Roger's (2003) Diffusion of Innovations explains the "pilot trap" not as a failure of technology, but as a failure of "compatibility" with existing hospital workflows and the Technology Acceptance Model (TAM) explains the psychological resistance that occurs in the healthcare workforce. Combining recent empirical evidence with these theoretical lenses, the review moves beyond a simple collection of feasibility studies to a critical examination of the structural and behavioural determinants of technology adoption.

Data Extraction Protocol

A standardised Data Extraction Form (DEF) (Appendix C) was developed in Excel to ensure transparent analysis across the 37 core papers. The extraction process was designed to divide findings into three theoretical domains of the research question.

1. **Study Metadata:** Author, Year, Location, and Journal.

2. **Contextual Factors:** Differentiates the Low-Resource Settings (e.g., Rwanda, Ghana) and High-Resource Settings (e.g., USA, Sweden) to analyse adoption barriers related to infrastructure.

3. **Payload & Focus:** Type of drone (Fixed wing vs. Multirotor) and payload such as Blood, Organs, AEDs, or Diagnostic Samples.

4. **Key Findings**

- *Managerial:* Extracted data related to cost-benefit analysis, regulatory hurdles, workforce training requirements, and organisational readiness.

- *Clinical:* Information regarding the stability of samples (haemolysis, temperature support) and clinical outcomes (such as reduced response times in OHCA cases)

- *Operational:* Technical performance metrics, including battery range, flight stability in extreme weather conditions, and successful delivery rates.

5. **Methodological Limitations:** The Author-identified limitations to explain the adaptability of findings.

By systematically populating this table, the review allowed a cross-case thematic analysis. This identified the recurring patterns such as operational fragility of drones in high-resource urban environments versus their success in rural low-resource environments, thus directly addressing the research question using detailed, comparable evidence sourced exclusively from the relevant literature.

Findings

This chapter presents a critical thematic synthesis on the adoption and effectiveness of drone-based medical transport. The analysis, based on a systematic review of 37 core empirical studies and nine foundational theoretical frameworks, progressing beyond a simple descriptive summary of technological capabilities to investigate the structural, clinical, and operational constraints in the field. The application of theoretical frameworks, particularly the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and Diffusion of Innovations (DOI) theory, is crucial for understanding the complex sociotechnical interactions that empirical data alone cannot fully explain. The literature presents a narrative of a rapidly advancing technology ready to transform healthcare logistics. However, a thorough analysis reveals a fragmented landscape characterised by methodological inconsistencies and significant asymmetries between theoretical potential and practical application.

Collectively, the 46 sources provide a solid foundation for identifying the mechanisms underlying the "pilot trap", the phenomenon in which successful small-scale trials fail to convert into systemic integration. The analysis reveals that drones have a theoretical potential to enhance transport in certain contexts; however, their effectiveness varies significantly based on environmental complexity, payload stability, and institutional maturity. A significant finding is the major difference in success factors according to location; the determinants for adoption in low-resource environments, where drones serve as substitutes for infrastructure, are fundamentally distinct from those in high-resource urban areas, where they face competition from established ground logistics. The following analysis synthesises these dimensions into three overarching themes: the sociotechnical divergence between strategic intent and

operational readiness; the validity gradient of clinical evidence across different biological payloads; and the discordance between simulation optimism and ecological reality.

Theme 1. The Sociotechnical Divergence: Strategic Intent Versus Organisational Readiness

Current research indicates that the primary challenge to adoption has shifted from technological limitations to institutional readiness. Many feasibility studies highlight the technical ability of drones to reduce delays in urgent medical transport, but the evidence shows that the necessary institutional conditions to maintain these benefits are often lacking. The existing literature shows that managerial adoption often moves beyond the availability of technology, as discussed in engineering-centric studies and it involves navigating through a complex sociotechnical environment that includes organisational divergence and regulatory uncertainty (De Silvestri et al., 2023; Daud et al., 2022).

A critical review of organisational readiness reveals a significant gap between the theoretical capabilities of the technology and the actual feasibility of its application. Managerial acceptance is primarily connected to the drone's perceived usefulness in minimising transport time and improving accessibility, a core construct of Davis's (1989) Technology Acceptance Model (TAM). This is supported by a cluster of high-profile case studies in the national-scale operations, such as Zipline's networks in Rwanda and Ghana, in addition to feasibility evaluations for blood and vaccine delivery in South Asia. These studies show time savings and enhanced **cold-chain**[2] reliability, supporting the theoretical notion that "relative advantage" is an important aspect in an organisation's intent to adopt (Dutta and Samantarai, 2024; Sayarshad, 2025; Gauba et al., 2025; Zailani et al., 2020) but the literature indicates that this decision should be addressed cautiously. Several substantial findings rely on narrow scoped

[2] **Cold-Chain –** The process of maintaining specific temperature for temperature-sensitive goods throughout the supply chain.

pilots conducted under favourable and non-replicable conditions. These trials are often funded and managed by drone manufacturers or innovation-oriented agencies thus guarding healthcare managers from the true operational costs (Damoah et al., 2021; Hiebert et al., 2020). This introduces a structural bias where managerial complexities such as cost structures, process redesign, insurance liability, and human resource planning are dismissed in favour of performance metrics like flight speed (Al-Alawi and AlEnezi, 2025). When managers are presented with data derived from these "sanitised" pilots, their intent to adopt is artificially inflated. They perceive the utility of the technology without perceiving the "ease of use" barriers that Davis (1989) identifies as critical for actual system usage. Consequently, the literature reveals a pattern where strategic interest is high during the pilot phase but collapses when the external support structures are removed, and the hospital must integrate the system into its routine budget and workflow.

Crucially, the literature reveals that managerial outlook exceeds organisational maturity. Leaders usually underestimate the operational complexity of integrating drones into existing supply chain and overestimate the staff readiness (World Health Organisation, 2022). For example, despite great enthusiasm for adoption at higher levels, front-line staff including nurses and porters remain unsure about necessary protocols such as payload handling, fault management, and coordination with EMS dispatch (Rees et al., 2021). This operational discordance is crucial as explained by the UTAUT (Venkatesh et al., 2003). While managers focus on "performance expectancy" (the idea that the drone will be faster), the workforce is primarily concerned with "effort expectancy" (the fear that the drone will make their job harder) and "facilitating conditions" (the availability of training and support). This analysis shows that the facilitators such as high-quality workforce training, liability, and system integration, are rarely incorporated in feasibility trails (Truog et al., 2020). Managers view the drone as a strategic asset whereas the workforce often views it as a source of disruptive

variability. Digital literacy barriers are low in high-resource hospitals, but workflow complexity is high since it requires the integration of several departments, like emergency room, blood bank, and laboratory, which creates friction points that are not present in low-resource settings. The transfer process from a drone to a sterile lab environment introduces new contamination measures that are not required in ground couriers. The lack of engagement with front-line staff presents a methodological weakness in the literature, which predominantly focuses on the perspectives of managers or drone operators rather than the end-users (Stierlin et al., 2024). As a result, managerial interest in short-term pilots often fades when the resource requirements for routine scaling arise, confirming the UTAUT hypothesis that without supportive facilitating conditions, usage behaviour will not be sustained (Zègre-Hemsey etal., 2024; Venkatesh et al., 2003).

This lack of readiness is further extended by regulatory and governance complexity. The literature identifies regulatory frameworks not just as a barrier, but as a defining constraint on managerial autonomy. The leading aviation authorities indicate that **Beyond Visual Line of Sight**[3] (BVLOS) abilities are highly necessary for medical use. A drone visible to the pilot offers no advantage over a conventional vehicle, yet these operations are governed by strict rules (Civil Aviation Authority, 2020; Federal Aviation Administration, 2022; International Civil Aviation Organisation, 2025). Numerous studies propose autonomous, long-distance drone networks, but they depend on limited exemptions that cannot be scaled (Poljak and Šterbenc, 2019). Managerial enthusiasm based on these exempted experiments is therefore unstable, as the regulatory framework required for ordinary, non-exempted operations does not yet exist in most jurisdictions (Quintanilla García et al., 2021). For instance, drone flights conducted during the COVID-19 pandemic response are operated under exceptional

[3] **BVLOS (Beyond Visual Line of Sight)** – An aviation term for operating a drone at distances where the pilot can no longer see the aircraft with the naked eye. This capability is essential for medical logistics to cover useful distances but requires advanced safety systems (Detect-and-Avoid) to be legal.

emergency feasibility permissions (Kumar et al., 2021), but those exceptions are rarely applied in standard healthcare delivery, indicating that management enthusiasm derived from emergency use outcomes may misrepresent long-term feasibility. Ethical governance, especially for high-risk payloads such as organs or infectious materials, receives even less systematic analysis. The High-profile cases of kidney transportation, often referenced in managerial discussions, represent single, heavily monitored events rather than scalable logistical frameworks (Scalea et al., 2021), and the lack of replication impacts the evidence necessary for strategic decision-making related to liability and risks (Kamberi et al., 2022). Until regulatory bodies provide a consistent framework for liability in the event of a drone crash involving a donor organ, managers in high-liability environments (US or UK) cannot rationally authorise routine adoption, regardless of the technological capability.

Finally, the theme highlights that managerial decision-making is heavily restricted by social validity. While technical studies often assume passive public acceptance, sociological research within the studies reveals considerable friction. Research focusing on the delivery of Automated External Defibrillators (AEDs) demonstrates this barrier clearly. There is considerable uncertainty among bystanders concerning their interactions with drones (Sedig et al., 2020). Participants in various mixed-methods studies reported discomfort regarding the proximity of autonomous aerial devices, noise, and privacy issues, which established behavioural barriers that directly challenge the managerial assumption that reduced arrival time correlates with clinical advantage (Zègre-Hemsey et al., 2020). If the public is unwilling to interact with the technology due to fear or confusion, the managerial value proposition diminishes (Sanfridsson et al., 2019). Rogers' (2003) Diffusion of Innovations framework states that adoption is affected by "compatibility" with pre-existing values and needs. According to the literature, drones can be viewed as "incompatible" with the public's expectation of quiet, secure airspace in high-resource urban settings. Until drones achieve

"social validity" where they are seen as a generous medical tool rather than a surveillance device or noise nuisance, managers in high-resource settings who face higher reputational risks will likely restrain their institutional intent for implementation (Rogers, 2003). This contradicts with low-resource settings, where the "relative advantage" of drones compared to vacant infrastructure is greater, resulting in reduced social scepticism, thus highlighting that social acceptance is a context-dependent variable (Dutta and Samantarai, 2024). In conclusion, Theme 1 indicates that the main barrier to adoption is not technological capability, but the absence of a supportive sociotechnical ecosystem. Regulatory fragmentation, staff unpreparedness, and societal reluctance negatively impact managerial enthusiasm, resulting in a gap between the intention to adopt and the ability to implement (Zailani et al., 2020; De Silvestri et al., 2023).

Theme 2. The Validity Gradient: Clinical Safety and Payload Integrity

A critical synthesis of the 37 empirical studies demonstrates that the quality of evidence is not uniform, and it degrades as the complexity of the payload increases. While the body of evidence provides robust support for the safety of stable medicines, it dangerously extends these findings to fragile and complex medical items without sufficient empirical support. This results in an "optimism bias," leading to the assumption that the established safety of blood transport is applicable to all medical logistics, despite the physiological differences in the payloads (Amukele et al., 2017).

The transport of robust biological medicines, particularly whole blood and standard diagnostic samples are at the strong end of the **validity gradient**[4]. The literature offers a high degree of consensus here. Empirical evaluations in various contexts indicate that core

[4] **Validity Gradient** – A conceptual term describing the spectrum of clinical evidence quality within the literature, which is robust for stable payloads (e.g., whole blood) but declines significantly for fragile biologicals (e.g., organs, monoclonal antibodies).

biochemical parameters, including haemolysis index, potassium concentration, and pH levels, exhibit no clinically significant differences between samples transported by drone and those transported by ground (Amukele et al., 2015; Amukele et al., 2017; Homier et al., 2020; Yakushiji et al., 2021). The consistent nature of these findings across various airframes (fixed-wing and multi-rotor), distances, and climates supports the claim that standard blood products show strong resistance to the mechanical stresses experienced during the flights (Homier et al., 2020; Yakushiji et al., 2021). This section of the literature is methodologically reliable, often using rigorous pre-flight and post-flight comparisons with control groups. However, the clinical review shows limitations in sample variety within the 'high validity' range of the curve. Several studies investigated small sample sizes containing lesser than twenty samples, thereby limiting the statistical power necessary to identify rare but clinically significant anomalies (Amukele et al., 2016; Stierlin et al., 2024). Moreover, many studies focus on standard blood bags (RBCs) or anticoagulated samples, resulting in a lack of evidence regarding more delicate components like platelets or cryoprecipitate, which may respond differently to the specific vibration spectra of UAVs (Gauba et al., 2025). Despite these minor limitations, the findings provide a strong conclusion that routine drone transport of diagnostic blood samples is clinically feasible.

As the analysis moves down the gradient to specialised and fragile payloads, the evidence becomes extremely weak. The transportation of solid organs is often presented in media and policy discussions as an innovative use of drone technology that may potentially solve the logistical issues in transplant medical care. The clinical evidence supporting this claim is still in preliminary stages. The literature predominantly depends on one prominent case study concerning kidney transplantation (Scalea et al., 2021). While this case demonstrated successful preservation parameters, the reliance on an "N=1" study design constitutes a major weakness for evidence-based policy (Kamberi et al., 2022). Unlike blood products, solid organs

are highly sensitive to vibration and temperature fluctuations. A single successful flight does not account for the variability of atmospheric conditions, turbulence profiles, or flight durations that a systemic network would encounter. Without multi-centre trials and data covering a variety of organ types (e.g., hearts or lungs, which are more sensitive to ischemia than kidneys), managerial claims regarding the systemic feasibility of organ drone transport remain speculative and scientifically premature (Scalea et al., 2021; Kamberi et al., 2022). This gap reveals the underlying problem that exists in the drone literature, which is the achievement of isolated successes is translated as the evidence of general feasibility, ignoring the high biological variability involved in organ transport. The current evidence proves possibility, not reliability.

The validity gradient decreases further when analysing complex biologicals, including monoclonal antibodies. Recent findings challenge the assumption of universal safety. Specific antibody formulations show considerable degradation (aggregation and fragmentation) when exposed to the rapid pressure changes and temperature differences linked to fluctuations in drone altitude (Güngören et al., 2024). This finding is important as it challenges the assumption that drones are a "plug-and-play" logistical solution for all medications. It suggests that clinical integration must be directed by payload-specific risk profiles instead of a broad acceptance of the technology (Güngören et al., 2024). Similarly, studies on plasma samples and coagulation tests reveal sensitivity to continuous vibration or thermal drift, which can modify clinical results (Yakushiji et al., 2021). The findings indicate that suitability is random upon specific payloads, and that clinical integration necessitates a more sophisticated approach to biochemical risk profiling than what is presently reflected in general feasibility studies. This evidence suggests that the physical stresses of drone flight specifically high-frequency vibration act as a pre-analytical variable that can compromise the integrity of complex proteins, a risk factor that is absent in stable blood samples.

The literature on vaccines complicates this theme by highlighting the gap between logistical success and clinical validation. Usually in low-resource settings, studies document clear logistical advantages like improvements in vaccine availability, reduction in stock-outs, and very few missed vaccination opportunities (Dutta and Samantarai, 2024; Zailani et al., 2020). However, a significant methodological weakness in many of these studies is the use of temperature logs as an alternative for clinical effectiveness (Gunaratne et al., 2024). Relatively limited research follows up with laboratory tests to ensure that the biological efficacy of the vaccine is preserved despite exposure to the vibration frequencies of drone flight. Although the temperature may be constant, micro-vibrations can compromise the efficiency of sensitive biological products (Gunaratne et al., 2024). As a result, the logistical case for vaccine drones is strong regarding access and speed, yet the clinical validation is still incomplete. The literature effectively proves that drones can keep vaccines cold, but it has not yet definitively proven that they keep vaccines effective under all flight conditions.

Finally, the literature on Automated External Defibrillator (AED) deployment reveals a significant gap between technical efficiency and clinical results. Simulation studies consistently report that drones can arrive at the scene of a cardiac arrest minutes faster than ambulances. In a situation where each minute of delay reduces survival probability by 7-10%, this is a clinically significant margin (Cheskes et al., 2020; Claesson et al., 2016; Rees et al., 2021). However, the current literature rarely addresses the human behavioural pattern required for converting arrival time into efficient defibrillation. Real-world and mixed-method studies reveal that bystanders often hesitate to retrieve or operate drone-delivered AEDs due to fear, confusion, or misunderstanding of instructions (Zègre-Hemsey et al., 2020; Sanfridsson et al., 2019). This behavioural barrier indicates that the operational success of rapid delivery does not necessarily result in the clinical success of early defibrillation. The clinical efficacy of drones in this context is not merely a technological characteristic; it is significantly influenced by

public education and user interface design, elements that are currently under-explored (Zègre-Hemsey et al., 2024). A drone that arrives in 3 minutes but is not used is clinically inferior to an ambulance that arrives in 10 minutes with trained paramedics. In conclusion, Theme 2 argues that clinical safety is not a binary characteristic of drone technology. The literature demonstrates high validity for robust payloads such as blood; however, it lacks equivalent evidence for organs, antibodies, and vaccines. The validity gradient creates clinical ambiguity, resulting in the managerial hesitation identified in Theme 1, which prevents the routine adoption of high-value medical goods.

Theme 3. Simulation Optimism Versus Ecological Reality

The operational domain is the most empirically measurable component, yet it is also most prone to "simulation optimism." A significant portion of the literature utilises mathematical modelling and simulation to demonstrate the speed, efficiency, and network potential of drones (Lakhwani and Sinjana, 2025; Lin et al., 2024). However, a detailed critique of these approaches indicates that they consistently exclude environmental volatility, resulting in conclusions that show high internal validity (the math is correct) but low external validity (the real world is more chaotic than the model) (Lakhwani et al., 2025).

Delivery time advantages, the most frequently referenced benefit of drones, are predominantly supported by simulated scenarios. Numerous studies utilise algorithms to demonstrate that drones significantly reduce response times by bypassing traffic congestion and terrain obstacles. While optimisation studies like Diego et al. (2024) predict linear efficiency gains, these findings stand in tension with empirical reports from Quintanilla García et al. (2021), which document non-linear failure modes when weather variables are introduced. Although mathematical models are reliable, they generally assume well-defined landing zones, direct line-of-sight dispatch, and minimal airspace congestion. Meteorological disruptions are

often excluded or minimised in these models (Rees et al., 2021). Empirical evidence from field reports indicates that mission delays or cancellations frequently occur due to factors such as wind gusts, precipitation, and battery degradation (Hiebert et al., 2020). This difference highlights the role of meteorological unpredictability as a critical limitation on operational reliability, a factor often neglected in theoretical optimisation models (Lakhwani and Sinjana, 2025). For example, the reliance on deterministic variables (average wind speed) rather than stochastic variables (gust profiles and precipitation probability) results in a systematic overestimation of fleet reliability in urban canyons. Although the theoretical potential for speed is clear, the achievable speed in a chaotic urban environment remains unproven.

This conflict is further amplified by the challenges of environmental sensitivity and payload vulnerability. Studies show that the drone platforms show structural sensitivity to weather conditions, impacting both reliability and clinical safety. Operational instability, particularly turbulence, acts as a logistical barrier and a clinical risk factor that can lead to sample degradation (Lammers et al., 2023). Usually, this instability increases in high-resource environments, where “urban canyons” formed by tall buildings cause signal interference and unstable wind shears (Ostermann et al., 2025). The evidence suggests that turbulence increases variability in biochemical markers, indicating that the “effectiveness” of transport is limited by the weather conditions (Lammers et al., 2023). The prominent real-world implementation of Zipline has taken place in rural, low-resource environments characterised by stable weather patterns and uninterrupted airspace (Dutta and Samantarai, 2024). This geographical difference indicates operational success currently relies on environmental simplicity rather than technology advancement. The literature often predicts success from rural Africa to urban Europe or North America, a transfer of findings that ignores the significant difference in airspace complexity (Quintanilla García et al., 2021). What actually works in the open skies of

Rwanda does not necessarily work in the congested airspace of London or New York, yet the literature often combines these contexts.

Furthermore, technical reliability and system failures are rarely documented. While industry narratives claim that drones are highly reliable, qualitative research reveals that employee anxieties arise regarding factors such as battery reliability, landing accuracy, and communication lag, which specifically affect operational predictability (Truog et al., 2020). These concerns are validated by reports from COVID-19 deployments, identifying network problems and GPS signal degradation as significant failure modes (Kumar et al., 2021). Despite this, many feasibility studies neglect the restrictions on airspace and the functionality of Global Navigation Satellite Systems (GNSS) (Civil Aviation Authority, 2020). The dependency on these idealised assumptions leads to misleading representation of readiness. Moreover, there is a lack of accurate data on failure rates. Most academic publications focus on successful missions, whereas aborted flights, deviations, and critical incidents are frequently excluded from datasets, leading to publication bias (Zègre-Hemsey et al., 2024). The lack of transparency prevents healthcare systems from accurately evaluating the risks linked to the integration of drones into critical care pathways, where reliability is highly crucial. If a drone carrying blood for a transfusion fails, the clinical consequence is severe; therefore, the reliability standard for medical drones must be exponentially higher than for commercial package delivery. The current literature focused on "proof of concept," has not yet provided the longitudinal failure-rate data necessary to establish this reliability.

Finally, the literature on supply chain integration indicates that simulation studies often depend on assumptions that contradict supply chain theory. Advanced models used for multi-drone coordination and routing predict major efficiency gains through AI integration (Damoah et al., 2021; Lin et al., 2024), but these simulation models usually assume ideal weather

conditions, unlimited charging capacity, and immediate deployment. The current established logistics theory indicates that the functional supply chains in unstable environments require resilience-oriented design such as redundancy and flexibility, rather than pure optimisation like speed and efficiency (Christopher, 2016; FAO and UNDP, 2020). The failure to integrate uncertainty, regulatory constraints, or behavioural variables in simulation research undermines the generalisability of their findings (Ostermann et al., 2025). Real-world deployments often face delays and infrastructural limitations that are often excluded in theoretical models. For instance, the duration needed to load a drone, conduct pre-flight safety checks, and obtain airspace clearance can negate the flight-time savings for short-distance flights. In summary, Theme 3 argues that the operational literature shows an ecological imbalance between mathematical possibilities and empirical feasibility. Drones have the capability for rapid delivery in controlled environments, but their real-world performance is heavily influenced by weather conditions, airspace complexity, and the maturity of the systems used. This "simulation optimism" results in a misleading perception of reliability when compared to the analysis of real-world logistics, thereby contributing to the managerial hesitation addressed in Theme 1. The data indicates that drones are not a universal solution; rather, they function as a niche tool most effective in predictable environments, hence challenging the current narrative of immediate, system-wide disruption.

Interdependencies and Implications

Managerial, clinical, and operational factors form a mutually constraining ecosystem. Managerial hesitancy, as explained in Theme 1, is not just risk aversion but a rational response to the "validity gradient" (Theme 2) and the "simulation optimism" (Theme 3) that form the existing evidence base (De Silvestri et al., 2023). Leaders cannot permit systemic scaling when operational reliability depends on ideal weather conditions or when clinical validation is

restricted to robust payloads like whole blood, leaving high-value biologicals at risk (Amukele et al., 2017; Lammers et al., 2023). This dependency suggests that the "sociotechnical divergence" is maintained by the disconnect between theoretical models and ecological reality (Lakhwani et al., 2025).

The main implication for the field is a transition from descriptive feasibility studies to robust implementation science. The findings indicate that healthcare administrators should broaden 'readiness' audits to include not only internal digital competence but also external ecological evaluations of local airspace complexity. To bridge the gap between pilot success and routine adoption, future research funding must prioritise longitudinal trials that explicitly measure failure rates, payload degradation in complex biologicals, and front-line workforce integration (Zègre-Hemsey et al., 2024; Stierlin et al., 2024). Furthermore, policy frameworks must evolve beyond temporary exemptions to establish harmonised performance-based navigation standards for BVLOS operations, rather than **blanket restrictions**[5] (CAA, 2020). Without addressing these structural interdependencies, drone logistics will remain confined to the "pilot trap," unable to transition from experimental novelty to a scalable, evidence-based standards.

[5] **Blanket restrictions** are rules or policies applied universally to all people or situations within a group without considering individual needs or risks. It follows a "one-size-fits-all" approach to managing risk, rather than tailoring rules for each person.

Discussion

This systematic review examined the managerial, clinical, and operational factors influencing the adoption of drone-based medical transport. This review presents a critical counter-narrative to the current "technological optimism" in the literature, which highlights the theoretical speed and reach of UAVs. By synthesising 37 empirical studies through sociotechnical theoretical lenses, this research advances the field by shifting the analytical focus from *technological feasibility* (can it fly?) to *systemic viability* (can it be sustained?).

Contribution to the Domain

The main contribution of this work is the identification of the "Validity Gradient" and the "Sociotechnical Schism." Previous reviews have viewed medical drone logistics as a homogenous domain, suggesting that effective blood delivery in Rwanda predicts successful organ delivery in New York (Dutta and Samantarai, 2024; Sayarshad, 2025; Gauba et al., 2025; Zailani et al., 2020). This review challenges that assumption. By categorising evidence based on payload fragility and environmental complexity, this work presents that the current evidence base is structurally uneven. It reveals that the "safety" of drone transport is not a universal constant but is dependent on the biological robustness of the payload. This difference is critical for decision-makers; it moves the discourse away from a binary "adopt/reject" decision toward a nuanced, risk-stratified approach to implementation. Furthermore, by applying the UTAUT and TAM frameworks, this review explains why the technology remains trapped in the "pilot phase", not because the technology fails, but because the institutional ecosystem (workforce, insurance, regulation) has not matured at the same pace as the airframes.

Future Directions and Research Agenda

The current state of the literature stands at a saturation point regarding feasibility studies. The field does not require further "proof-of-concept" flights for standard blood products in clear weather; that validity is established. Instead, to bridge the gap between experimental novelty and routine standard of care, future research must focus on Implementation Science.

1. **Transitioning from Simulation to Stress Testing:** In the future, operational studies must shift from "simulation optimism". Research designs must explicitly integrate unpredictable weather variables, including wind gusts and precipitation, as well as urban interference, to generate realistic reliability data.

2. **From N=1 to Clinical Trials:** Clinical research must move beyond single-case organ flights. Large-scale, multi-centre trials are required to establish the safety profile for fragile biologics (monoclonal antibodies, live organs) under routine flight conditions.

3. **Workforce Integration:** Managerial research must focus on the "absorptive capacity" of healthcare workers. Longitudinal studies are required to measure how drone integration impacts the daily workflow of nurses and lab technicians, moving beyond the viewpoint of the drone operators.

Conclusion

Integrating drones into healthcare logistics represents a paradigm shift capable of overcoming the rigidities of terrestrial infrastructure. This review concludes the sector is defined by a sociotechnical divergence, where strategic ambition exceeds operational readiness.

The evidence confirms that drones are a clinically valid solution for robust payloads like whole blood, particularly in low-resource settings where they act as infrastructure substitutes. However, for fragile payloads and high-resource urban environments, the "optimism gap" remains a critical risk factor. The dependence on simulations that ignore ecological reality and clinical studies with limited sample sizes has created a premature sense of preparedness. For drone logistics to transition from a niche solution to a systemic standard of care, the focus must shift from proving that drones can fly, to proving that they can fit into the chaotic, regulated, and cautious reality of modern healthcare. Until the "validity gradient" is compressed through rigorous, high-volume testing of fragile payloads, widespread adoption remains a strategic aspiration rather than an operational reality.

Learning Statement

Framework: The "What?" Model by John Driscoll

The process of this research and development has been a cognitive shift from considering technology as a standalone solution to recognising it as a complex part of a larger sociotechnical system. Using Driscoll's (1994) reflective framework, I have outlined my learning journey of this development process into three phases: the original undertaking (What), the transformation of my perspective (So What), and applying this knowledge and skills in the future (Now What). This reflective approach structured around Driscoll's framework navigates through the stages of Kolb's (1984) cycle, determining how I translated Concrete Experience of research challenges into the Active Experimentation of a new professional identity.

What? (The Context and Challenge)

Initially, I approached this review with a linear perspective, expecting that the barriers to be primarily technical (battery density, payload capacity, and flight range). My objective was to categorise these factors along with managerial resistance. However, data extraction revealed a major intellectual challenge, the fragmentation of the literature. Engineering papers discussed vibration frequencies (operational) without referencing hospital liability (managerial), while management papers discussed "resistance to change" without acknowledging the valid clinical reasons for that resistance, such as haemolysis risks. The primary difficulty was the integration of these distinct "themes." I found myself struggling to synthesise variables that did not naturally sit together, attempting to correlate wind shear statistics with the psychological constructs of the Technology Acceptance Model (TAM).

So What? (The Shift in Perspective and Skill Development)

This challenge led to the most profound shift in my critical perspective. I realised that treating managerial, clinical, and operational factors as separate "silos" was not just a structural choice, but a critical flaw in my initial thinking. I learned that managerial hesitancy is not merely "resistance," but a rational response to operational ambiguity, a realisation that resulted the concept of the "Validity Gradient" in my findings. This cognitive restructuring highly resembles the Abstract Conceptualisation phase of Kolb's (1984) experiential learning cycle, where I moved beyond the concrete experience of fragmented data to synthesise a new, unified theoretical model.

Successfully navigating this integration required me to develop interdisciplinary skillset. I had to become fluent in interpreting clinical analysis (to understand payload degradation) while simultaneously applying sociological frameworks like UTAUT to workforce behaviour. This process moved me from being a passive consumer of information to an active critic of methodology. I learnt to look past the "simulation optimism" of engineering research to question the ecological validity of their claims. This shift was transformative as I stopped asking "Can the drone fly?" and started asking "Can the system incorporate this technology?". This move from technological optimism to systematic reality was the defining arc of my skill development.

Now What? (Future Application and Professional Identity)

The insights learned from this review have changed my approach to innovation management. I came to realise that successful implementation is not just dependent on advancement of the hardware, but the level of maturity within the receiving ecosystem. Moving forward, I will apply the "sociotechnical" perspective developed in this review to ensure that when introducing new technologies, I am not only solving for efficiency but solving for legitimacy and

integration. This research has provided me with the necessary persistence to look for the "validity gradient" in any evidence base to ensure that decisions are informed by solid reality rather than theoretical potential. Ultimately, this work has transformed me from a student of logistics to an analyst in implementation science.

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Appendices

Appendix A

MMAT 2018 Format (Excel Template)

Ref ID	First author	Year	Citation

①

	SCREENING QUESTIONS	
②	S1. Are there clear research questions?	S2. Do the collected data allow to address the research questions?

1. QUALITATIVE STUDIES				
1.1. Is the qualitative approach appropriate to answer the research question?	1.2. Are the qualitative data collection methods adequate to address the research question?	1.3. Are the findings adequately derived from the data?	1.4. Is the interpretation of results sufficiently substantiated by data?	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?

③

	2. RANDOMIZED CONTROLLED TRIALS			
④	2.1. Is randomization appropriately performed?	2.2. Are the groups comparable at baseline?	2.3. Are there complete outcome data?	2.4. Are outcome assessors blinded to the intervention provided?	2.5 Did the participants adhere to the assigned intervention?

3. NON-RANDOMIZED STUDIES				
3.1. Are the participants representative of the target population?	3.2. Are measurements appropriate regarding both the outcome and intervention (or exposure)?	3.3. Are there complete outcome data?	3.4. Are the confounders accounted for in the design and analysis?	3.5. During the study period, is the intervention administered (or exposure occurred) as intended?

⑤

4. QUANTITATIVE DESCRIPTIVE STUDIES				
4.1. Is the sampling strategy relevant to address the research question?	4.2. Is the sample representative of the target population?	4.3. Are the measurements appropriate?	4.4. Is the risk of nonresponse bias low?	4.5. Is the statistical analysis appropriate to answer the research question?

⑥

5. MIXED METHODS STUDIES				
5.1. Is there an adequate rationale for using a mixed methods design to address the research question?	5.2. Are the different components of the study effectively integrated to answer the research question?	5.3. Are the outputs of the integration of qualitative and quantitative components adequately interpreted?	5.4. Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?	5.5. Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?

⑦

COMMENTS

⑧

Appendix B

References rated using MMAT 2018 tool (Based on above format)

Source No.	Reference	MMAT Category	Screening (S1, S2)	Criteria Met (e.g., 4.1-4.5)	Quality Rating	Notes (Rationale from Findings)
1	Al-Alawi & AlEnezi (2025)	4. Quant Descriptive	Yes	4.1, 4.2, 4.3	*** (Med)	Theoretical model; lacks longitudinal validation.
2	Amukele et al. (2016)	4. Quant Descriptive	Yes	All Met	***** (High)	Rigorous pre/post-flight haemolysis validation.
3	Amukele et al. (2017)	4. Quant Descriptive	Yes	All Met	***** (High)	Long-distance validation; robust sample size.
4	Amukele et al. (2015)	4. Quant Descriptive	Yes	All Met	***** (High)	Foundational proof-of-concept; clear variables.
5	Cheskes et al. (2020)	4. Quant Descriptive	Yes	4.1, 4.3	*** (Med)	Simulation assumes ideal conditions (optimism gap).
6	Claesson et al. (2017)	2. Quantitative RCT (Randomised Controlled Trial)	Yes	2.1, 2.2, 2.3	***** (High)	Strong comparative design (Drone vs EMS).
7	Claesson et al. (2016)	4. Quant Descriptive	Yes	4.1, 4.2	*** (Med)	GIS model; theoretical time savings only.

8	Damoah et al. (2021)	5. Mixed Methods	Yes	5.1, 5.2	*** (Med)	Case study focus; limited generalisability.
9	Daud et al. (2022)	1. Qualitative	Yes	1.1, 1.2	*** (Med)	Scoping review; synthesises disaster applications.
10	De Silvestri et al. (2023)	1. Qualitative	Yes	1.1, 1.3	*** (Med)	Review of barriers; high relevance but secondary data.
11	Diego et al. (2024)	4. Quant Descriptive	Yes	4.1, 4.5	*** (Med)	Optimisation model; assumes ideal data flow.
12	Dutta & Samantarai (2024)	5. Mixed Methods	Yes	All Met	***** (High)	National-scale case study (Zipline); high ecological validity.
13	Gauba et al. (2025)	5. Mixed Methods	Yes	5.1, 5.2	*** (Med)	Feasibility study; specific to Indian context.
14	Gunaratne et al. (2024)	4. Quant Descriptive	Yes	4.1, 4.2	*** (Med)	Simulation model; limited by theoretical inputs.
15	Güngören et al. (2024)	4. Quant Descriptive	Yes	All Met	***** (High)	Lab experiment; identifies specific failure modes (antibodies).
16	Hiebert et al. (2020)	1. Qualitative	Yes	1.1, 1.4	*** (Med)	Review focused on North America; context specific.
17	Homier et al. (2020)	2. Quantitative RCT (Randomised Controlled Trial)	Yes	All Met	***** (High)	Urban pilot (Montreal); robust control group.
18	Kamberi et al. (2022)	4. Quant Descriptive	Yes	All Met	***** (High)	Pre-clinical organ transport; strong telemetry data.
19	Kumar et al. (2021)	4. Quant Descriptive	Yes	4.1	*** (Med)	System design proposal; theoretical resilience.

20	Lakhwani & Sinjana (2025)	4. Quant Descriptive	Yes	4.1, 4.5	*** (Med)	Algorithm/Math model; lacks field validation.
21	Lakhwani et al. (2025)	4. Quant Descriptive	Yes	4.1, 4.5	*** (Med)	Optimisation algorithm; theoretical efficiency.
22	Lammers et al. (2023)	5. Mixed Methods	Yes	5.1, 5.3	*** (Med)	Combat zone context; specialized but valid.
23	Lin et al. (2024)	4. Quant Descriptive	Yes	4.1, 4.5	*** (Med)	Fleet optimisation model; assumes perfect coordination.
24	Ostermann et al. (2025)	1. Qualitative	Yes	1.1, 1.4	*** (Med)	Systematic review; secondary data analysis.
25	Poljak & Šterbenc (2019)	1. Qualitative	Yes	1.1, 1.3	*** (Med)	Microbiology review; highlights regulatory gaps.
26	Quintanilla García (2021)	5. Mixed Methods	Yes	5.1, 5.4	***** (High)	Real-world COVID deployment; high operational validity.
27	Rees et al. (2021)	4. Quant Descriptive	Yes	4.1, 4.2	*** (Med)	Monte Carlo simulation; valuable but theoretical.
28	Sanfridsson et al. (2019)	5. Mixed Methods	Yes	All Met	***** (High)	Combines simulated delivery with bystander interviews.
29	Sayarshad (2025)	4. Quant Descriptive	Yes	4.1, 4.5	*** (Med)	Equity model; theoretical optimisation.
30	Scalea et al. (2021)	4. Quant Descriptive	Yes	4.3, 4.4	*** (Med)	N=1 Case study; high validity but low generalisability.
31	Sedig et al. (2020)	1. Qualitative	Yes	All Met	***** (High)	Focus groups; captures public perception barriers.
32	Stierlin et al. (2024)	3. Quant non-Rand	Yes	3.1, 3.2	***** (High)	Comparative study (Drone vs Courier); robust data.

33	Truog et al. (2020)	1. Qualitative	Yes	All Met	***** (High)	In-depth community interviews; strong credibility.
34	Yakushiji et al. (2021)	4. Quant Descriptive	Yes	All Met	***** (High)	Vibration monitoring experiment; strong sample integrity data.
35	Zailani et al. (2020)	1. Qualitative	Yes	1.1, 1.2	*** (Med)	Maternal health review; limited empirical trial data.
36	Zègre-Hemsey et al. (2024)	5. Mixed Methods	Yes	5.1, 5.3	***** (High)	Cross-national comparative case study; high relevance.
37	Zègre-Hemsey et al. (2020)	4. Quant Descriptive	Yes	4.1, 4.3	***** (High)	User experience simulation: strong human-factor focus.

Appendix C

Data Extraction Form (DEF)

Source No.	Reference	Context / Focus LR = Low-Resource Settings, HR = High-Resource Settings	Detailed Payload / Focus	Study Design Type	Specific Content / Focus	Perspective	Key Findings	Limitations / Notes
1	Al-Alawi & AlEnezi (2025)	LR/Remote: General medical consumables and urgent supplies.	Medical Supplies: General medical consumables and urgent supplies.	Pilot/Model	Effectiveness investigation via theoretical modelling.	Operational	Highlights potential for reaching underserved areas where infrastructure is lacking.	Lacks longitudinal data on routine integration.
2	Amukele et al. (2016)	HR (USA): Blood Products (RBCs, Platelets, Plasma).	Blood Products: RBCs, Platelets, and Plasma (3 units each).	Validation Study	Experimental flight trial comparing pre/post-flight haemolysis.	Clinical	Validated stability of blood products; no significant haemolysis found comparing pre/post-flight.	Short duration flight; controlled conditions.
3	Amukele et al. (2017)	HR (USA): Diagnostic Samples (Chemistry and Haematology).	Diagnostic Samples: Chemistry and Haematology (84 pairs).	Flight Trial	Long-distance (3-hour) flight validation over arid terrain.	Clinical	Long-distance flight (up to 3 hrs) over arid terrain did not compromise sample integrity.	Small sample batch; favourable weather.

4	Amukele et al. (2015)	HR (USA): Coagulation, Chemistry, and Haematology specimens.	Diagnostic Samples: Coagulation, Chemistry, and Haematology specimens.	Proof-of-Concept	Pilot study assessing vibration impact on analytes.	Clinical	Established early proof-of-concept for routine lab specimen transport impact on analytes.	Early-stage feasibility only.
5	Cheskes et al. (2020)	HR (Canada): Automated External Defibrillator (AED).	Medical Device: Automated External Defibrillator (AED).	Feasibility Study	Comparative timed trials (Drone vs. Ambulance).	Operational / Clinical	Drones arrived faster than EMS in rural settings; potential to reduce time-to-defibrillation.	Simulations often assume ideal landing zones.
6	Claesson et al. (2017)	HR (Sweden): AED (750g payload).	Medical Device: AED (750g payload).	Randomised Controlled Trial (RCT) - Simulation	Randomised flight trials vs. EMS in simulated cardiac arrest scenarios.	Operational / Clinical	Drone arrived 16 mins faster than EMS; high potential for survival increases in OHCA.	Excluded bad weather/no-fly zones.
7	Claesson et al. (2016)	HR (Sweden): AED.	Medical Device: AED.	GIS Analysis	Geographic Information System modelling of historical cardiac arrest data.	Operational	GIS modelling showed theoretical time savings for OHCA coverage.	Theoretical model; no physical flight data.
8	Damoah et al. (2021)	LR (Ghana): Supply Chain AI integration.	Supply Chain: AI integration for sustainability and efficiency.	Case Study	Qualitative analysis of AI-enhanced drone logistics.	Managerial	AI integration can optimise fleet management for sustainability and efficiency.	Lacks empirical validation in clinical workflow.

9	Daud et al. (2022)	Global: Disaster Relief (Blood, Medicines, Search & Rescue).	Disaster Relief: Blood, Medicines, and Search & Rescue tech.	Scoping Review	Synthesis of drone applications in disaster management.	Managerial	Drones are useful in disaster relief where infrastructure is broken.	Focus is on disaster response, not routine care.
10	De Silvestri et al. (2023)	HR/Global: Routine Logistics (Lab samples and pharma).	Routine Logistics: Lab samples and pharmaceuticals.	Scoping Review	Analysis of barriers to routine (non-emergency) implementation.	Managerial	Identifies regulatory and cost barriers to “routine” (non-emergency) application.	Synthesis of existing barriers rather than new data.
11	Diego et al. (2024)	Global: Disrupted Supply Chains (Internet of Drones).	Internet of Drones (IoD): Information flow and physical logistics.	Optimization Model	Mathematical modelling of resilience under disruption.	Operational	Improved resilience of drone networks under supply chain disruption.	Theoretical optimisation model.
12	Dutta & Samantarai (2024)	LR (Rwanda): Blood & Vaccines (Zipline).	Blood & Vaccines: Zipline distribution network.	Case Study	Analysis of Zipline’s hub-and-spoke operational model.	Managerial	Zipline hub-and-spoke model demonstrates national-scale feasibility and access equity.	Success is specific to Rwanda’s regulatory environment.
13	Gauba et al. (2025)	LR (India): Blood Delivery (Time-critical).	Blood Delivery: Time-critical blood units.	Feasibility Study	Evaluation of efficiency and sustainability metrics.	Managerial / Operational	Evaluation of efficiency and sustainability metrics for blood delivery.	Focus on feasibility metrics rather than clinical outcomes.
14	Gunaratne et al. (2024)	LR (Sri Lanka): Vaccines (Cold chain).	Vaccines: Cold-chain distribution.	Simulation	District-based optimisation model for vaccine coverage.	Operational	District-based approach improves cold-chain efficiency.	Model-heavy; limited long-term implementation data.

15	Güngören et al. (2024)	HR: Biologics (Monoclonal Antibodies).	Biologics: Monoclonal Antibodies (mAb) formulations.	Lab Experiment	Stress-testing stability against drone-specific vibration/shock profiles.	Clinical	Some antibodies degrade under pressure/temp changes; contradicts “universal safety” claims.	Highlights the need for payload-specific validation.
16	Hiebert et al. (2020)	HR (North America): General Health Services.	General Health Services: Transport and remote sensing.	Scoping Review	Overview of applications and environmental barriers.	Operational	Wind gusts/turbulence frequently cause delays or mission cancellations.	Highlights environmental sensitivity often ignored.
17	Homier et al. (2020)	HR (Canada): Blood Products (Simulated).	Blood Products: Simulated blood payloads.	Randomised Controlled Trial (RCT) - Pilot	Randomised comparison of drone vs. ground delivery times in urban traffic.	Operational / Clinical	Faster than ground in urban setting; no damage to blood products.	Specific atmospheric conditions required.
18	Kamberi et al. (2022)	HR (USA): Organs (Donor Kidney).	Organs: Donor Kidney (pre-clinical).	Experimental	Telemetry and biopsy analysis of porcine/human kidneys post-flight.	Clinical	Pre-clinical validation of organ transport parameters; biopsy analysis of porcine/human kidneys.	Pre-cursor to human trials; controlled setting.
19	Kumar et al. (2021)	Global: COVID-19 Supplies (Contactless delivery).	COVID-19 Supplies: Test kits, sanitisers, and samples.	System Architecture	Proposal for a networked drone system for contactless delivery.	Operational	Proposal for networked drone systems for pandemic response/contactless delivery.	Conceptual system design.

20	Lakhwani & Sinjana (2025)	General: Logistics Networks (Routing).	Routing: Healthcare supply chain logistics.	Metaheuristic Model	"Erosion-Enhanced Rain Optimisation" algorithm for routing.	Operational	"Erosion-Enhanced Rain Optimisation" algorithm optimises healthcare supply routes.	The mathematical model assumes ideal flight conditions.
21	Lakhwani et al. (2025)	General: Multimodal Logistics (Blood Bags).	Blood Bags: Temperature-sensitive routing.	Mathematical Model	Optimisation algorithm for multimodal (drone + ground) transport.	Operational	"Rain Optimisation Algorithm" improves multimodal (drone + ground) transport efficiency.	Theoretical efficiency; lacks real-world weather data.
22	Lammers et al. (2023)	Conflict Zone: Blood & Plasma (Battlefield).	Blood & Plasma: Battlefield logistics (Whole Blood).	Review/Feasibility	Analysis of UAV utility in combat casualty care.	Operational	Feasible in combat zones ("Airborne" study) but turbulence affects markers.	Military context may not apply to civilian safety standards.
23	Lin et al. (2024)	HR: Emergency Services (Heterogeneous payloads).	Emergency Items: Heterogeneous payload assignment.	Optimization Model	Algorithm for task assignment across a mixed drone fleet.	Operational	Algorithm optimizes task assignment across a mixed drone fleet.	Simulation assumes perfect fleet coordination.
24	Ostermann et al. (2025)	General: Automated Logistics (Network Planning).	Network Planning: System performance metrics.	Systematic Review	Synthesis of planning approaches for logistic networks.	Managerial	Synthesis of planning approaches and system performance metrics.	Secondary data analysis.
25	Poljak & Šterbenc (2019)	Global: Clinical Labs (Infectious Samples).	Infectious Samples: Microbiology specimens (Biohazard Cat B).	Review	Current status and regulatory barriers for infectious substances.	Clinical	Review of drones for infectious disease sample	Notes regulatory barriers for infectious substances.

							transport; notes regulatory barriers.	
26	Quintanilla García (2021)	HR (Spain): Critical Goods (COVID-19).	Critical Goods: COVID-19 samples and supplies.	Case Study	Real-world deployment analysis during a healthcare system stress period.	Operational	Real-world use case in Valencia showing quick deployment during stress periods.	Real deployments face delays absent in simulations.
27	Rees et al. (2021)	HR (UK): Medical Device (AED).	Medical Device: AED.	Computer Simulation	Monte Carlo simulation of response times (Drone vs. Ambulance).	Operational	Monte Carlo simulation of response times (Drone vs. Ambulance).	Assumed airspace availability.
28	Sanfridsson et al. (2019)	HR (Sweden): Medical Device (AED).	Medical Device: AED.	Mixed Methods	Simulated delivery followed by semi-structured interviews with bystanders.	Managerial / Social	Bystanders hesitated to use drone-delivered AEDs (technical speed = clinical success).	Highlights behavioural barriers often overlooked.
29	Sayarshad (2025)	LR: Vaccines (Equity-based distribution).	Vaccines: Equity-based distribution.	Mathematical Model	Optimisation for equitable coverage in resource-poor settings.	Managerial	“Equity-based” delivery optimisation for disease-prone regions.	Focus on equity metrics; theoretical model.
30	Scalea et al. (2021)	HR (USA): Organs (Human Kidney).	Organs: Human Kidney for transplant.	Case Study (N=1)	First-in-human real-world drone delivery of an organ.	Clinical / Managerial	First successful human organ delivery with normal biopsy; high resource/regulatory burden.	N=1 (Single case); heavily monitored.

31	Sedig et al. (2020)	HR (Canada): Medical Device (AED).	Medical Device: AED.	Qualitative	Focus groups and interviews on public perception.	Managerial / Social	Public perception analysis: identified “What is an AED?” confusion.	Underscores public education gap.
32	Stierlin et al. (2024)	HR (Switzerland): Diagnostic Samples (Blood chemistry).	Diagnostic Samples: Blood chemistry.	Comparative Study	Matched pairs analysis of sample integrity (Drone vs. Courier).	Managerial	Challenges for “routine” use identified; regulatory and workflow misalignment.	Highlights the gap between pilots and routine integration.
33	Truog et al. (2020)	LR (Malawi): General Medical (Community perception).	General Medical: Community perception of drone operations.	Qualitative	Interviews and focus groups (“Insights Before Flights”).	Managerial / Social	“Insights Before Flights”; community perceptions and acceptance determine success.	Community acceptance is a “go/no-go” factor.
34	Yakushiji et al. (2021)	HR (Japan): Blood Products (RBCs).	Blood Products: Red Blood Cell (RBC) solutions.	Experimental	Quality control analysis following inter-island flight.	Clinical / Operational	RBCs maintained quality during remote island transport; vibration monitoring included.	Small sample size; specific geographic context.
35	Zailani et al. (2020)	LR (Malaysia): Maternal Care (Oxytocin/Blood).	Maternal Care: Oxytocin and blood products.	Review	Feasibility of drones for maternal mortality reduction.	Clinical	Potential for maternal healthcare products to reduce mortality in remote areas.	Review of potential rather than new trial data.
36	Zègre-Hemsey et al. (2024)	HR (USA, Sweden, Canada): Medical Device (AED).	Medical Device: AED.	Comparative Case	Cross-national analysis of integration barriers.	Managerial	Lessons from cross-national analysis on barriers to real-time integration.	Comparative policy analysis.

37	Zègre-Hemsey et al. (2020)	HR (USA): Medical Device (AED).	Medical Device: AED.	User Experience (UX)	Simulation focused on human-drone interaction (bystander usability).	Operational / Social	User experience analysis of human-drone interaction during simulated arrest.	Human-drone interaction friction points identified.